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Simulation-based multi-objective optimization combined with a DHM tool for occupant packaging design

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ABSTRACT

Occupant packaging design is usually done using computer-aided design (CAD) and digital human modelling (DHM) tools. These tools help engineers and designers explore and identify vehicle cabin configurations that meet accommodation targets. However, studies indicate that current working methods are complicated and iterative, leading to time-consuming design procedures and reduced investigations of the solution space, in turn meaning that successful design solutions may not be discovered. This paper investigates potential advantages and challenges in using an automated simulation-based multi-objective optimization (SBMOO) method combined with a DHM tool to improve the occupant packaging design process. Specifically, the paper studies how SBMOO using a genetic algorithm can address challenges introduced by human anthropometric and postural variability in occupant packaging design. The investigation focuses on a fabricated design scenario involving the spatial location of the seat and steering wheel, as well as seat angle, taking into account ergonomics objectives and constraints for various end-users. The study indicates that the SBMOO-based method can improve effectiveness and aid designers in considering human variability in the occupant packaging design process.

1. Introduction

In vehicle design, occupant packaging is a layout problem involving selecting locations for components such as pedals, seats, and steering wheel in the vehicle cabin under certain geometric constraints, e.g., general vehicle cabin dimensions. A general aim of occupant packaging processes is to identify vehicle cabin configurations that meet defined accommodation levels for expected vehicle occupants, or to optimize accommodation levels when targeted levels cannot be met due to conflicting requirements. Generally, accommodation measures the number of individuals who can find a comfortable posture to perform all required transportation tasks (Roe, 1993; Gkikas, 2016; Bubb et al., 2021). Component location is critical because it is the primary constraint on occupant postures, affecting both satisfaction and safety (Scott et al., 1996; Parkinson and Reed, 2008; Smith et al., 2015; McMurry et al., 2018; Reed et al., 2019; Gao et al., 2022; Jeong et al., 2023). Considering a diverse population with different body types and behavioural preferences is imperative to ensure a good fit, safety, comfort, and accessibility for targeted end-users. However, diversity consideration introduces one of the most significant challenges in occupant packaging, often requiring design engineers and ergonomics designers to consider several interdependent and sometimes conflicting factors and objectives, such as addressing roominess and reachability for people of different sizes while maintaining the vehicle's safety, aerodynamics, and overall styling (Wolf et al., 2022). The consideration of interdependent and conflicting factors occurs throughout the development process via balancing activities until the design is fully defined. For that, designers aim to find the best balance ("compromise") that enhances the vehicle's performance without sacrificing the ergonomics and safety of individuals with varying anthropometrics (Bhise, 2016; Kikumoto et al., 2021). The balancing activities involve various tools or methods, including physical prototypes, user tests and interviews, computer-aided design (CAD), and digital human modelling (DHM) tools to investigate or simulate the interaction between the occupants and the vehicle interior (Gkikas, 2016; Perez Luque et al., 2022a). Although CAD and DHM tools have significantly advanced the design process by enabling design and simulations in a virtual world, the use of such tools still involves issues, such as time-consuming procedures and occasionally providing questionable simulation results, which in turn, can lead to a need for costly re-design iterations or validation studies (Wolf et al., 2020; Parida et al., 2020; Demirel et al., 2021).

Previous studies identify that ergonomics designers typically follow a

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standard procedure when using DHM tools in occupant packaging design (Green, 2000; Hanson et al., 2006, 2008; Gkikas, 2016; Perez Luque et al., 2022b). The standard procedure typically involves two main activities: 1) The digital manikin(s) is (are) set to an initial posture, typically a default posture based on predefined body joint angles; 2) Constraints are then defined to ensure basic requirements are fulfilled, e.g., placing feet on the pedals and hands on the steering wheel. Depending on the DHM tool, the sequence of these steps may vary, but typically, the essential activities remain the same. Next, the DHM tool automatically positions the manikin(s) in a posture that is considered optimal, i.e., the most likely seated posture according to the DHM tool's prediction method, while fulfilling the defined constraints. To represent anthropometric diversity, the DHM standard procedure is generally performed using a set of manikins, collectively often referred to as a *manikin family*, where each manikin has a systematically defined combination of anthropometric measurements (Brolin et al., 2019).

Since occupant packaging design is a complex act of finding a successful balance between many kinds of requirements, there is a need for methods that can assist design engineers and ergonomics designers in considering multiple objectives and constraints in a simultaneous optimization process. Simulation-based multi-objective optimization (SBMOO) is a process that combines digital simulations with optimization methods, with the aim of finding superior solutions to design problems involving multiple, potentially conflicting, objectives. In the SBMOO process, the design problem is modelled in a simulation tool, and relevant conflicting factors are defined as objective functions (Carson and Maria, 1997; Lorig, 2019). Recent studies have shown the potential of SBMOO in manufacturing contexts (Barrera Diaz et al., 2021; Guo et al., 2021; Lind et al., 2023). In regards to vehicle design, SBMOO has been applied to performance and component's design (Xu et al., 2004; Nandi et al., 2015; Holjevac et al., 2020; Astaneh et al., 2022; Wang et al., 2024) as well as vehicle structure design from a safety point of view (Gu et al., 2013; Dai et al., 2023; Gao et al., 2024). However, the application of SBMOO methods to occupant packaging design is still relatively unexplored. More traditional optimization techniques have been applied to solve vehicle occupant packaging problems, in which optimization techniques are used as a stochastic posturing method to consider anthropometric and postural variability (Parkinson and Reed, 2006; Parkinson et al., 2007). However, these approaches do not incorporate optimization techniques combined with DHM simulations. SBMOO has recently been combined with DHM tools for production design contexts (Iriondo Pascual et al., 2021). The framework in Iriondo Pascual et al. (2021) for performing SBMOO consists of three main phases: 1) Problem definition and creation of the optimization model; 2) Optimization process; and 3) Presentation and selection of results.

This paper investigates the potential advantages and challenges of using a SBMOO method combined with a DHM tool in occupant packaging design, following the framework in Iriondo Pascual et al. (2021). The aim is to investigate and discuss how such an approach can improve the effectiveness of the occupant packaging design process, addressing two main challenges common in current process: limited support for balancing multiple and conflicting objectives, and the tendency to perform limited analyses due to time-consuming procedures.

2. Method

This section describes how Phases 1 and 2 in the SBMOO framework in Iriondo Pascual et al. (2021) were set up and run to solve a fabricated occupant packaging design scenario.

2.1. Problem definition and creation of the optimization model

2.1.1. Problem definition

One crucial aspect concerning driver accommodation in occupant packaging is selecting the appropriate size and position of the seat adjustment range. The issue of sizing and placing adjustment ranges becomes more complicated for ergonomics designers if both the seat and the steering wheel are adjustable. Very large adjustment ranges for all components would accommodate almost all drivers. However, adjustability ranges are typically limited by other design considerations, including costs, space, and safety. Thus, the seating adjustment range specification is often a critical step for ergonomics designers in occupant packaging design.

A design scenario was created to investigate the application of SBMOO in occupant packaging design. The scenario involved determining the location of the seat adjustment range centre, the position of steering wheel, and the angle of the seat bottom. To achieve this, the location specification of these elements was divided into five variables: fore-aft and vertical location of the seat adjustment range centre, fore-aft and vertical location of the steering wheel centre, and angle of the seat bottom. These constitute the optimization *variables* in the design scenario (black arrows in Fig. 1). The variables have a limited range within which they can be optimized (grey areas in Fig. 1). The optimization problem is constrained by vehicle geometries such as the dimensions and locations of the roof, pedals, dashboard, and windscreen. The seat adjustment range, which has a fixed size area, is also part of the constraints (light blue delimited area in Fig. 1). A set of manikins with different anthropometric measurements were considered, each having unique optimal joint angles for a seated driving posture (clarified in Section 2.2.3). In this way, each manikin's combination of optimal body joint angles represents one *objective* of the optimization problem.

The design scenario illustrates an optimization problem in which comfort-related postural aspects (manikin's combination of body joint angles) are considered when optimizing the seat adjustment range centre position, steering wheel position, and seat bottom angle. In addition, four design constraints are added to the optimization problem (red arrows in Fig. 1): head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle, which are measured and evaluated against predefined targets along with the comfort-related postural aspects.

2.1.2. Model creation (DHM tool)

The DHM scene (Fig. 2) was modelled in a research version of the DHM tool IPS IMMA version 2023.R2 (IPS IMMA, 2023). A family of 14 manikins (7 female, 7 male), each with different anthropometrics (Table 1), was defined using the CAESAR data set (Robinette et al., 2002). Using existing manikin definition functionality in IPS IMMA, anthropometric data for each manikin was generated from two 3D boundary ellipsoids (one ellipsoid per sex) with a confidence level of 90% (Brolin et al., 2012), i.e. assuming a 90% accommodation level for the targeted population given by the CEASAR data set. Three body dimensions were selected as key dimensions: stature, body mass, and sitting height. This selection was based on the body dimensions used in the regression equations in Park et al. (2016). In addition to central cases (Manikins 1 and 2 in Table 1), representing an "average" manikin per sex, six boundary cases were defined at the ends of the three axes of each of the two ellipsoids, each representing test subjects with extreme but realistic combinations of the key anthropometric variables (Manikins 3 through 14 in Table 1). In IPS IMMA version 2023.R2, anthropometric data, including weight, is treated as normally distributed when defining the multidimensional ellipsoids and boundary cases. However, weight is positively skewed in the CAESAR data set and therefore the percentile values reported here, calculated based on standardised Z-values, are not identical to the actual percentile values of the CAESAR population.

An original Volkswagen Beetle (Type 1) vehicle CAD model was used to represent a relatively small cabin size (Fig. 2) and to provide a generic environment to test and demonstrate the SBMOO approach. This CAD model can be found in the training repository of Siemens Jack (Siemens, 2017). The origin (0,0) of the scene, consisting of the CAD geometry imported in the DHM tool, is located in the centre of the vehicle in x direction and on ground level in z direction (Fig. 2). Measurements in x

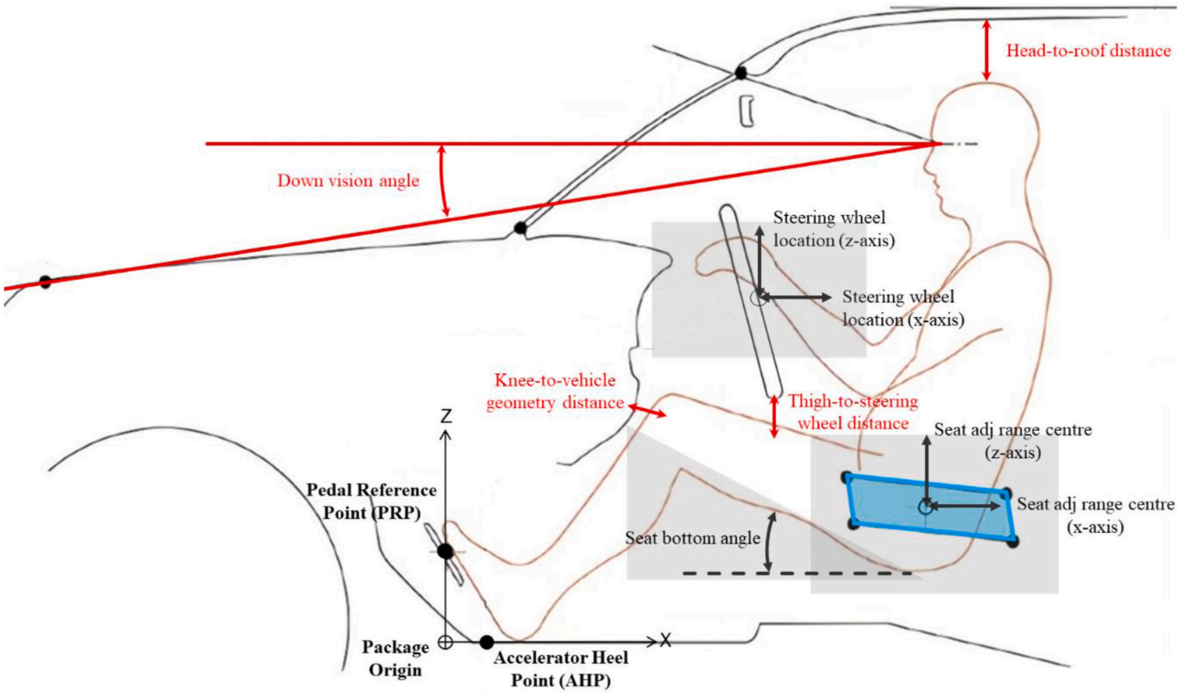

Fig. 1. Optimization definition: variables (black arrows); variables' range limit (grey areas); fixed size area of seat adjustment range (light blue); design constraints (red). Image based on Parkinson and Reed (2006). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 1
Manikin family anthropometric measurements.

Manikin	Sex	Stature	Z	Percentile	Sitting height	Z	Percentile	Weight	Z	Percentile
1	F	1639	0.000	50.0	865	0.000	50.0	66	0.000	50.0
2	M	1776	0.000	50.0	925	0.000	50.0	85	0.000	50.0
3	F	1601	−0.535	29.6	839	−0.732	23.2	94	2.028	97.9
4	M	1732	−0.563	28.7	892	−0.836	20.2	114	1.732	95.8
5	F	1694	0.757	77.6	838	−0.759	22.4	65	−0.096	46.2
6	M	1833	0.718	76.4	898	−0.686	24.6	83	−0.132	44.7
7	F	1806	2.315	99.0	947	2.257	98.8	85	1.368	91.4
8	M	1960	2.326	99.0	1016	2.260	98.8	114	1.732	95.8
9	F	1678	0.535	70.4	891	0.708	76.0	38	−2.073	1.9
10	M	1819	0.540	70.5	957	0.787	78.5	55	−1.817	3.5
11	F	1584	−0.772	22.0	892	0.736	76.9	67	0.050	52.0
12	M	1719	−0.727	23.4	952	0.663	74.6	86	0.048	51.9
13	F	1472	−2.330	1.0	783	−2.284	1.1	47	−1.414	7.9
14	M	1592	−2.338	1.0	834	−2.284	1.1	56	−1.757	4.0

Note: F = female; M = male; Stature in mm; Sitting height in mm; Weight in kg.

and z direction from the Scene origin to the Packaging origin, are given in Fig. 2. The Body Shape Alignment approach in Perez Luque et al. (2022b) was followed to make manikins adopt an initial seated driving posture within the virtual environment. Six so called attachment points were used in IPS IMMA to align manikin meshes to a virtual template of the H-point machine (SAE J826, 2021; SAE J4002, 2022) in addition to positioning the manikins' hands on the steering wheel, and right foot on the accelerator. These attachment points limit the potential movements of the manikins to a two or three-dimensional defined range (a line, a plane, or a box). Fig. 3a illustrates the six attachment points (T12L1 surface, T6T7 surface, lower back surface, buttock bottom surface, and right and left knee crease) used to align the manikin mesh to the H-point machine template, referring some of the attachment points to internal joints and others to mesh surfaces (see Perez Luque et al., 2022b for more details). Once the attachment points are defined, the DHM tool defines a final posture following a quasi-static posture prediction method for each manikin that fulfils the defined attachment points (Bohlin et al., 2012; Delfs et al., 2013).

2.2. *Optimization process*

2.2.1. *Optimization method*

Candidate design variants are identified through an iterative process of simulation and optimization, where each design variant consists of a different seat adjustment range centre, steering wheel position, and seat bottom angle. For a given design variant, simulations produce data for objective functions, which are used to evaluate the current design variant (clarified in Section 2.2.3). The genetic optimization algorithm NSGA-II was selected for its efficiency in multi-objective optimizations, its suitability for handling a range of correlated objectives, and the ability to incorporate objectives and constraints of both integer and real number nature in optimizations (Deb et al., 2002). An initial random population of 50 design variants was generated within the constraints of the vehicle cabin to seed the NSGA-II algorithm. Then, iteratively, the algorithm evaluated the data obtained from each simulation from the previous population, and then generated a new child population. The NSGA-II algorithm specifies variable values for new design variants by

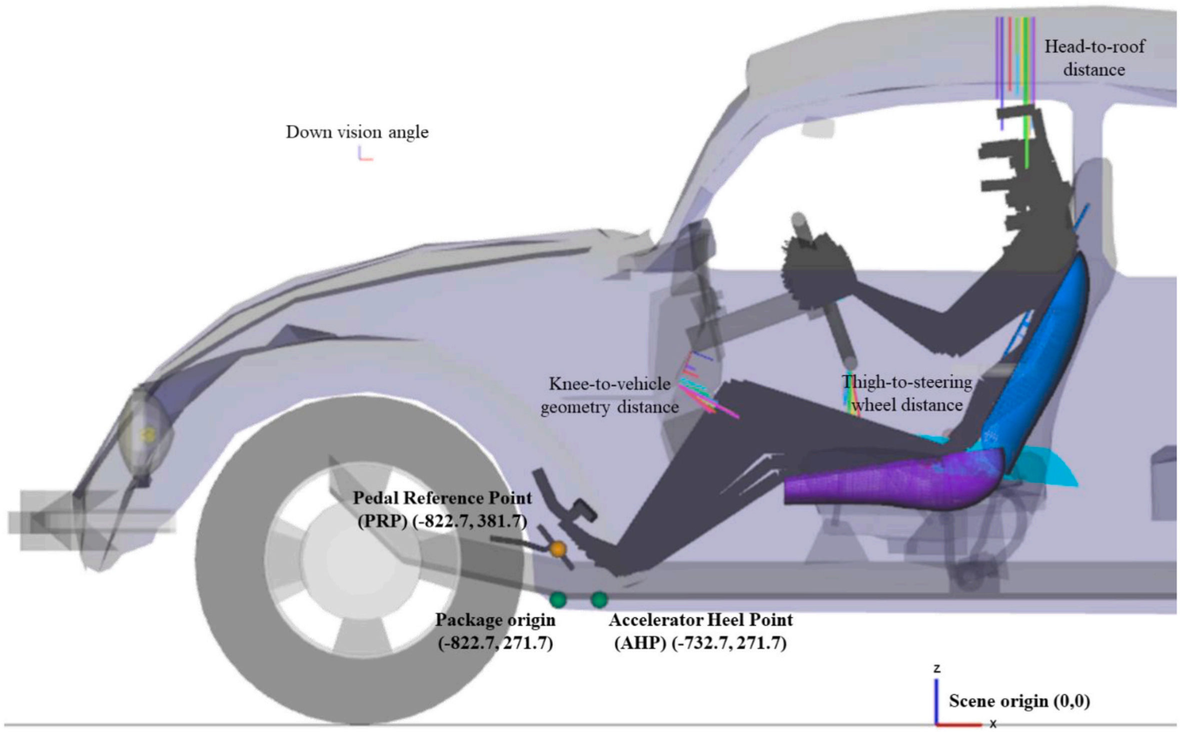

Fig. 2. Scene in the DHM tool: vehicle geometries, manikin family seated in a driving position (stick figure visualisation), and design constraints (head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle). Measurements in mm. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

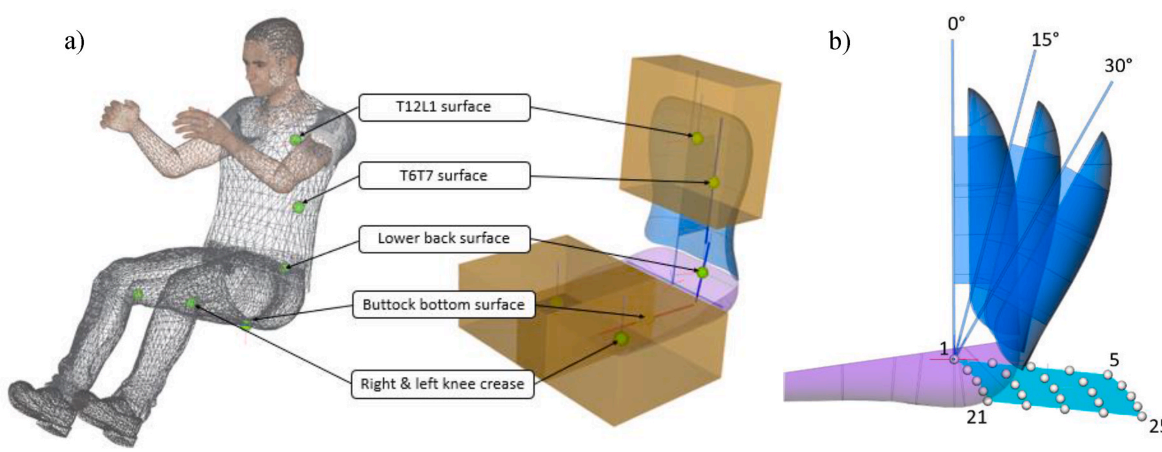

Fig. 3. a) Attachment points of the H-point template for the Body Shape Alignment approach; b) 25 seat adjustments (white spheres) inside the seat adjustment range (light blue shape) and three torso angles (0, 15, and 30°). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

applying crossover, mutation and selection operators (Deb et al., 2002) (Table 2). NSGA-II continued this process for 1000 iterations. Each design variant consisted of a different seat adjustment range centre, steering wheel position, and bottom seat angle. All 14 manikins were simulated for each design variant, including new values of seat adjustment range centre, steering wheel position, and bottom angle, using a multi-simulation approach (described in Section 2.2.2).

Table 2
Optimization method configuration.

Optimization method	NSGA-II algorithm
Population size	50
Child population size	50
Crossover probability	0.95
Mutation probability	0.05
Tournament size	2
Maximum iterations	1000

2.2.2. *Simulation method*

As described, each design variant was evaluated by simulation in the DHM tool, and then the evaluation results were provided to the NSGA-II algorithm to continue the optimization process. Each design variant, with a defined seat adjustment range centre, steering wheel position, and seat bottom angle, provides many possible configurations for vehicle occupants to set the seat position and back angle within the seat adjustment range, which consequently affects their driving posture. In order to simulate these possible configurations within the adjustment ranges in a DHM tool, 25 equidistant seat positions covering the entire seat adjustment range (white spheres in Fig. 3b) and three torso angles (0, 15, and 30° in Fig. 3b) were specified. The seat adjustments (25) combined with the torso angles (3) were considered in each design variant, resulting in 75 sitting postures per manikin (14) in each design variant.

A multi-simulation approach was used to evaluate the 75 sitting postures, running multiple sub-simulations for each of the 75 sitting postures for the 14 manikins. Therefore, 1050 sub-simulations conform to a complete simulation for one design variant. The simulation method step in the optimization framework in Iriondo Pascual et al. (2021) can be represented as a flowchart of sub-steps to perform the sub-simulations (Fig. 4). Once the design variant was automatically set up in the DHM tool, the first seat adjustment of the seat adjustment range was set. For each seat adjustment, each manikin was automatically seated using three torso angles (0, 15, and 30°).

Simulation values of each manikin and sub-simulation were stored for later analysis. Iteratively, results from the 1050 sub-simulations were evaluated in the next steps of the optimization process. This simulation method step (Fig. 4) was performed for every design variant to generate evaluation data to feed the optimization algorithm.

2.2.3. Driver posture evaluation

As the basis for quantifying comfort and inclusion, joint angle values of the manikins simulated in the DHM tool were compared with joint angle values from the regression models in Park et al. (2016). Based on sex, age, anthropometrics, and vehicle measurements, these regression models predict hip-to-eye vector, head, neck, thorax, abdomen, pelvis, right thigh, and right knee angles (Park et al., 2016). To explain posture variation, Park et al. (2016) give the root mean squared error (RMSE) for each joint angle. In this study, these RMSE values were used to specify an acceptable range of deviation between the manikin joint angles in the simulations and those predicted by the regression models (pRMSE). A simulated joint angle was considered acceptable ("OK") if the difference between the simulated and the model predicted value was not greater than pRMSE for that joint. If a joint angle deviation was greater than its corresponding pRMSE, the joint angle was considered unacceptable ("NOT OK"). Thus, joint angle acceptability was defined for each joint of each manikin individually in terms of an acceptable range as opposed to an optimal value. The number of acceptable and unacceptable joint angles were counted to quantify the overall acceptability of a manikin posture, providing an objective function to evaluate driver postures for the design variants generated in the optimization process.

There was a joint angle offset between the skeleton definitions provided by IPS IMMA (Hanson et al., 2019) and used in this study, and the skeleton definitions provided by the HumanShape framework (UMTRI HumanShape, 2020) and used in the Park et al. (2016) study. As a result, joint angles provided by IPS IMMA could not be directly compared to the results which were used in the body angle equations for posture evaluation. This body joint angle offset was compensated by measuring and incorporating the differences between the skeleton definitions in the equations for the posture evaluation. Offset values for the compensation were based on differences measured between joint angles for a neutral posture with arms resting at the side (n-pose). Table 3 lists the identified and incorporated offsets for each manikin and joint, including the hip-to-eye, head, neck, thorax, abdomen, pelvis, thigh, and knee.

2.2.4. SBMOO definition: mathematical modelling

In the mathematical definition of the design scenario, manikin anthropometries and postures were considered in the optimization objectives along with the design constraints. The design constraints were specified as mandatory minimum distances between the manikin and the vehicle interior, which are typically considered in occupant packaging design. In this case, these included: head-to-roof distance, knee-to-vehicle geometry distance, thigh-to-steering wheel distance, and down vision angle (Fig. 2). There were 14 optimization objectives, i.e. one evaluation of total number of unacceptable angles per manikin (numbers of "NOT OK"). The indices and parameters of the design scenario, with the addition of outputs to calculate the number of design constraint violations, are given in Table 4, and measurement limits of the constraints, with assumed realistic limit values, are given in Table 5.

The total number of unacceptable joint angles for each manikin in each seat adjustment for one design variant (f_{m_t}) was determined by calculating the absolute difference between the angle value from the simulated manikin and the predicted angle for the manikin and comparing it to the pRMSE (r) in Park et al. (2016) for a specific joint angle i and anthropometrics of a manikin m.

$$f_{m_t} = \sum_{i=1}^{I} \left[\left| a_{im_t} - p_{im_t} \right| > r_{im} \right] \quad (1)$$

f_{m_t} represents the number of angles that deviate more than r_i from the regression model predicted value for manikin m in adjustment t. Prac-

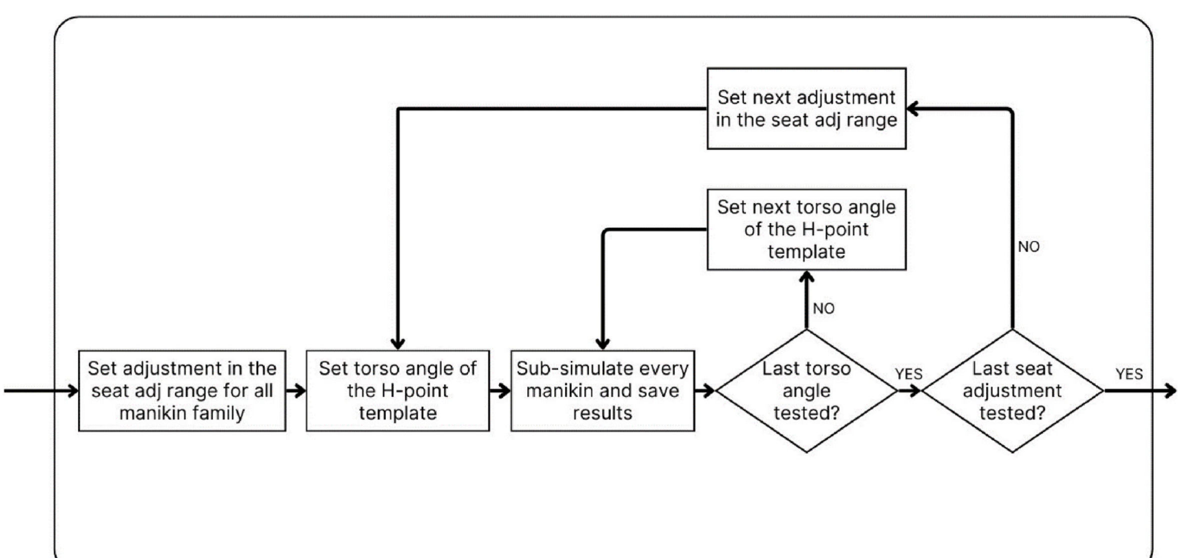

Fig. 4. Multi-simulation approach with sub-simulations performed for each occupant packaging design variant.

Table 3
Skeleton offset per manikin and joint angle (degrees).

Manikin	Hip-to-eye	Head	Neck	Thorax	Abdomen	Pelvis	Thigh	Knee
1	4.16	−27.06	−11.43	4.21	−0.36	8.1	2.79	−8.59
2	7.31	−24.28	−3.26	9.89	−2.22	14.04	5.25	−10.63
3	2.12	−29.09	−14.04	−0.52	4.92	6.68	0.82	−6.27
4	5.5	−25.64	−7.64	6.01	2.3	12.87	2.71	−9.64
5	3.92	−27.14	−10.99	1.95	4.32	7.78	0.86	−7.48
6	6.68	−24.17	−3.7	7.71	0.57	13.63	2.83	−7.94
7	−0.44	−32.51	−17.87	−5.36	1.33	4.22	−1.06	−3.59
8	3.1	−28.45	−11.25	1.3	4.27	10.47	0.87	−9.1
9	1.6	−30.45	−15.27	−1.47	2.39	5.42	0.99	−7.33
10	4.99	−26.74	−6.97	5.03	0.4	11.94	3.67	−11.68
11	0.14	−31.43	−17.8	−4.15	4.09	4.44	0.81	−5.78
12	3.79	−28.02	−10.86	2.88	2.92	10.61	2.57	−11.92
13	1.83	−29.72	−14.59	−1.05	3.64	5.87	0.71	−6.56
14	5.2	−26.24	−7.36	5.48	1.26	12.34	3.1	−10.86

Table 4
Indices, parameters, and simulation outputs of the design scenario.

Indices	Parameters	Simulation outputs
$i = 1.I$ Angle index	$I = 8$ Total number of body joint angles	$a_{i_{m_t}}$ Simulated body joint angle values from manikin
$m = 1.M$ Manikin index	$M = 14$ Total number of manikins	$p_{i_{m_t}}$ Predicted body joint angle values from manikin
$j = 1.J$ Design constraint index	$J = 4$ Total number of design constraints	$c_{j_{m_t}}$ Design constraint measurement
$t = 1.T$ Adjustment index	$T = 75$ Total number of adjustments	
	r_{i_m} pRMSE of each body joint angle of manikin	

Table 5
Constraint limits of the design scenario.

Design constraints name	Design constraint limit
Head-to-roof distance (mm)	$l_1 = 50$
Knee-to-vehicle geometry distance (mm)	$l_2 = 50$
Thigh-to-steering wheel distance (mm)	$l_3 = 50$
Down vision angle (°)	$l_4 = 13$

tically, f_{m_t} represents the number of unacceptable joint angles for a given manikin for each seat and torso angle adjustment. The solutions were also evaluated for the design constraints violated. For each manikin, m, and adjustment, t, a violation was identified when the measurement value of the design constraint c_j was less than the design constraint limit l_j. The total number of violations h_{m_t} was calculated:

$$h_{m_t} = \sum_{j=1}^{J} \left[l_j \geq c_{j_{m_t}} \right] \tag{2}$$

The complete set of adjustments evaluations, S, including the violated constraints and the optimization objectives for each manikin m was defined:

$$S_m = \left\{ h_{m_t}, f_{m_t} \right\} \tag{3}$$

The result of constraints violated for each manikin was calculated as the minimum value of h_m^{Min} in S_m:

$$h_m^{Min} = Min\{h_{m_t} | m_t \in S_m\} \tag{4}$$

Then, to calculate the number of unacceptable angles, the solution with the minimum unacceptable angles, f_m^{Min}, was selected from the solutions that fulfilled $S_m = h_m^{Min}$:

$$f_m^{Min} = Min\left\{ f_{m_t} | m_t \in \left\{ S_m = h_m^{Min} \right\} \right\} \tag{5}$$

The resulting minimum constraints violated of each manikin, h_m^{Min}, and minimum unacceptable angles of each manikin f_m^{Min} were given to the optimization algorithm to evaluate the design variant.

3. Results

This section describes results of the SBMOO method, which corresponds to Phase 3 in the framework in Iriondo Pascual et al. (2021).

3.1. Presentation of results

During the optimization process, 1000 design variants were iterated (generated), with each iteration performing 1050 sub-simulations. Convergence, which is the point where the iterative process stabilises by producing design variants that no longer significantly improve the specified objectives, was reached at 550 iterations (i.e., design variants generated). The number of objectives in the design scenario was 14, therefore, the Pareto front had 14 dimensions. Due to the complexity of data representation for a solution space of 14 dimensions, additional metrics showing single values as results of each iteration/design solution were used to analyse the results.

Fig. 5 shows the decision space (value limits for the optimization variables) and the solution space (each grey line represents a design variant of the occupant packaging design problem) using a parallel coordinate plot chart. In a parallel coordinate plot, each design variant corresponds to a line and each column represents a different optimization variable. These lines intersect with the vertical axes, one for each variable, and the position of each line along these axes conveys the value of the corresponding variable. In Figs. 5 and 6, the columns represent the five optimization variables and their corresponding value limits to represent the design space.

Next, the results of the design variants can be filtered to facilitate the solution selection process. In Fig. 6, filters are applied to highlight design variants that meet predefined criteria to represent the solution space. This is to illustrate how data filters can support decision processes, where colours are used to highlight variants that meet the filter criteria. The defined filter criteria include solutions that: have the minimum number of unacceptable angles for the entire manikin family (orange), have the minimum number of unacceptable angles per individual manikin (blue), and fulfil both filters (magenta). The first filter helps identify variants that have the best overall outcomes. The second filter helps identify the worst individual outcome for a given design variant. The third filter helps identify the best design variants' results considering the overall and individual outcomes (first and second filters combined). Different filters can be particularly useful as they allow identifying variants where results vary at an overall or individual level.

First, design variants that violated the design constraints were eliminated, and then the filters were applied. The filter value for the maximum unacceptable angles for the entire manikin family (orange) was specified as 56, equivalent to 4 unacceptable angles per manikin,

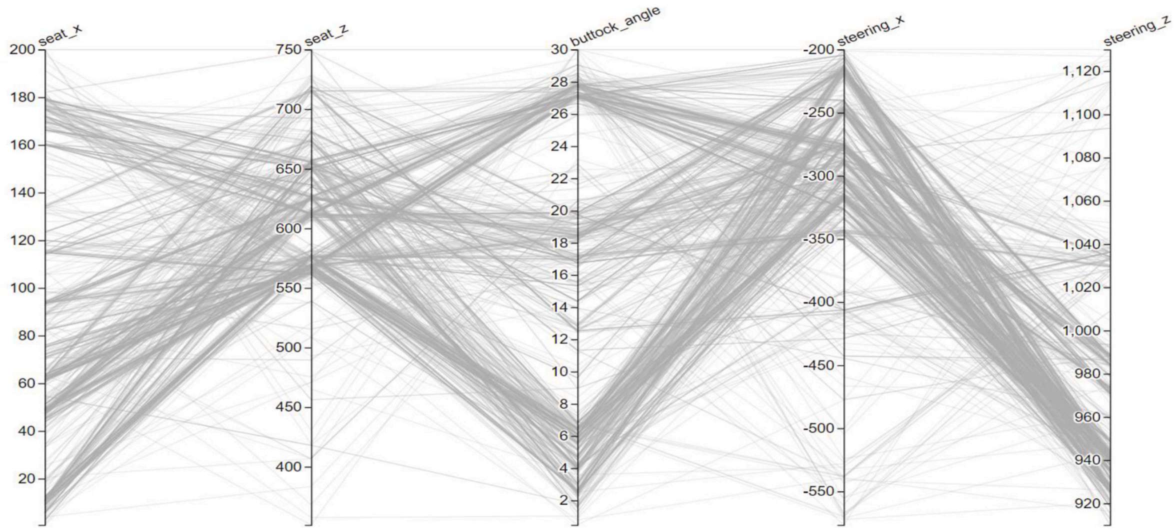

Fig. 5. Design and solution space without filters.

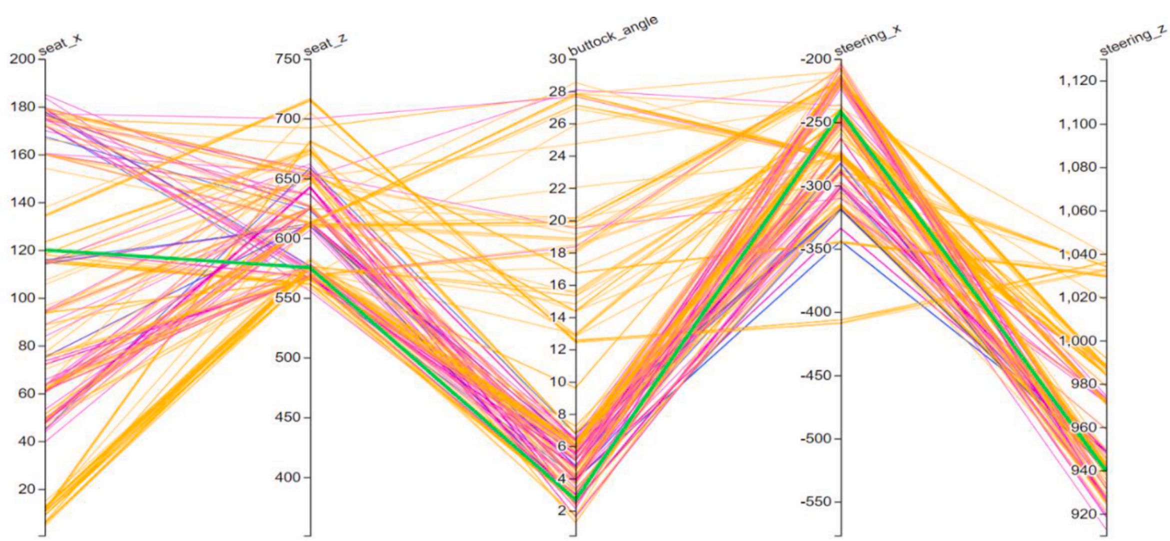

Fig. 6. Filtered solution space: max of unacceptable angles for manikin family = 56 (orange); max of unacceptable angles per individual manikin = 5 (blue); solutions fulfilling both filters (magenta); selected design variant for analysis (green). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

and the filter value for the maximum unacceptable angles per individual manikin (blue) was specified as 5.

Fig. 7 illustrates the filtered solution in 2D plots representing a side view of the cabin as in Fig. 2, in which the seat adjustment range centre and the steering wheel position are visible following the same colour filtering as in Fig. 6. Each design variant is represented by two points in the plot, one representing the steering wheel position (top left cluster of points) and another representing the seat adjustment range centre (bottom right cluster of points) in the x- and z-axis. Regarding both the seat adjustment range centre and the steering wheel position, design variants fulfilling both filters (highlighted in magenta) are located in specific areas and not along all the tested areas across both dimensions. The steering wheel solutions are low along the z-axis, favouring an increased visual angle for drivers, and the seat adjustment centre is often farther back along the x-axis to avoid a knee collision with the vehicle geometry. This pattern is likely caused by the design constraints in the design scenario, particularly the down vision angle and the knee-to-vehicle geometry distance.

3.2. Selection of solution

In practice, an ergonomics designer may want to select specific design variants for more detailed examination. To illustrate this process, one design variant of the design scenario was chosen as an example following the third filter and focusing on the minimum number of unacceptable angles at the overall and individual level. More detailed information about the design variant and simulation results are given in Table 6. The selected design variant is highlighted in green in Figs. 6 and 7.

Table 7 corresponds to an in-depth analysis of the postures of the 14 manikins for the selected design variant, describing the adjustment index (or indices) of the seat adjustment range and torso angle value(s) providing the least unacceptable angles in the chosen design variant. In this case, the number of violated constraints is 0, but it is possible to include design variants with constraint violations at this stage if detailed information about those violations is of use. Fig. 8 illustrates the posture of manikins 1 and 2 in the adjustment index 16(15°) and 21(30°) of the selected design variant.

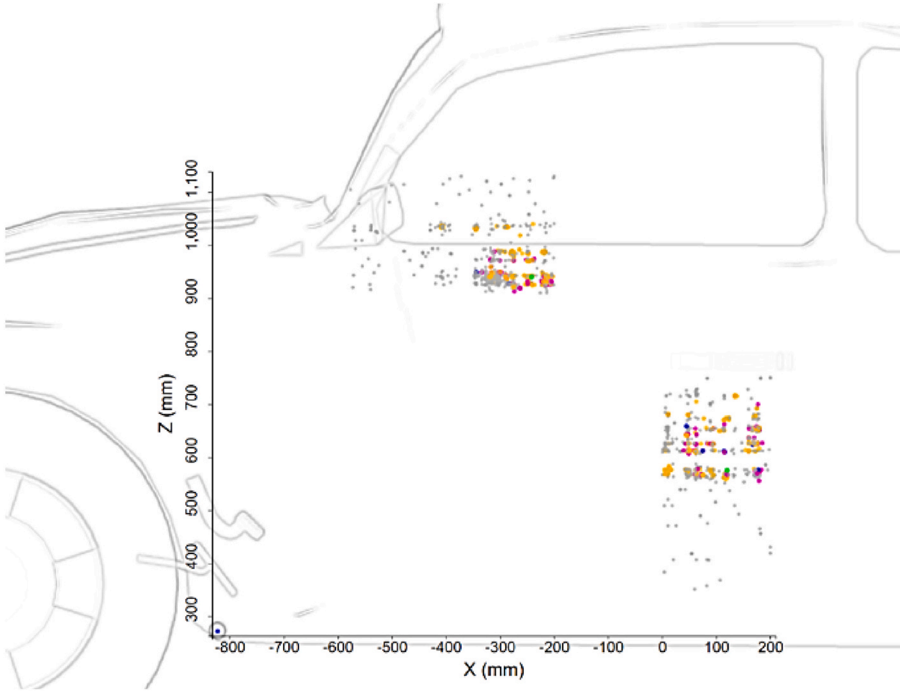

Fig. 7. Filtered solutions for seat adjustment range centre (bottom right cluster of points) and steering wheel position (top left cluster of points). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 6
Design information for selected design variant fulfilling both filters.

	Design information
Solution iteration/Design variant	#659
Seat adjustment range centre location in x-axis (mm)	120
Seat adjustment range centre location in z-axis (mm)	575.3
Steering wheel location in x-axis (mm)	−241.7
Steering wheel location in z-axis (mm)	939.8
Seat bottom angle (°)	2.7
The total sum of unacceptable angles per manikin family	38
Maximum unacceptable angles per manikin	4

Table 7
Manikin posture in-depth analysis for the selected design variant.

Manikin	Best seat adjustment index (1–25) & torso angle (0°/15°/30°) (fewest unacceptable angles)	No. of unacceptable joint angles	Names of unacceptable joint angles for the first best seat adjustment index and torso angle
1	16(15°), 21(15°)	2	hip-to-eye, neck
2	21(30°)	2	neck, pelvis
3	5(0°), 5(15°), 5(30°), 10 (0°), 10(15°), 10(30°), 15(0°), 15(15°), 15 (30°), 16(15°), 21(15°)	4	hip-to-eye, thorax, thigh, knee
4	21(30°)	3	hip-to-eye, neck, thorax
5	16(15°)	2	hip-to-eye, neck
6	5(30°), 9(30°)	3	neck, thigh, knee
7	17(15°), 21(15°), 22 (15°)	3	hip-to-eye, neck, thorax
8	22(30°)	2	neck, pelvis
9	21(15°)	1	abdomen
10	5(30°)	3	abdomen, thigh, knee
11	8(0°)	4	neck, abdomen, thigh, knee
12	16(30°), 21(30°)	3	hip-to-eye, neck, thorax
13	2(15°), 4(0°), 4(15°), 4 (30°)	3	neck, thigh, knee
14	7(30°)	3	neck, thigh, knee

4. Discussion

Occupant packaging activities are crucial for optimizing vehicle cabin space, ensuring comfort and safety for a wide range of occupants. The proposed SBMOO method combines DHM simulation and multi-objective optimization, using a genetic algorithm, to automatically generate and evaluate design solutions to find a successful one. The study was done to explore how SBMOO can enhance the effectiveness of occupant packaging design, addressing issues such as limited support for conflicting objectives and the tendency to limit analyses due to time constraints.

Firstly, the SBMOO method shows, when applied in the design scenario, how it can assist ergonomics designers to identify occupant packaging design solutions that balance multiple objectives while reducing constraint violations. This departs from the conventional, less capable, single-objective optimization approach commonly used in occupant packaging. Secondly, the SBMOO method makes it possible to consider variability across a wide range of dimensions (manikin, optimization variables, etc.) by automatically running many simulations to consider possible variations and providing interpretable results to ergonomics designers. As a result, SBMOO enables the consideration of targeted end-users with more granularity, for example, by considering the optimization of each manikin as its own objective. The ability to automatically evaluate the design space also minimises the subjectivity in the simulation process, improving simulation traceability and repeatability. In this way, design engineers and ergonomics designers can rather focus their expertise and experience on evaluation and decision-making processes, potentially increasing the effectiveness of the occupant packaging design process. Depending on the design scenarios, various features, objectives, and constraints can be tested and different evaluation methods can be deployed while facilitating the consideration of human anthropometric and postural variability in driving postures, enriching the design process.

The study illustrates the application of a framework introduced in Iriondo Pascual et al. (2021) and evaluates the results relative to the statistical prediction models specified by Park et al. (2016) based on a fabricated design scenario. Although the design scenario is fabricated, it

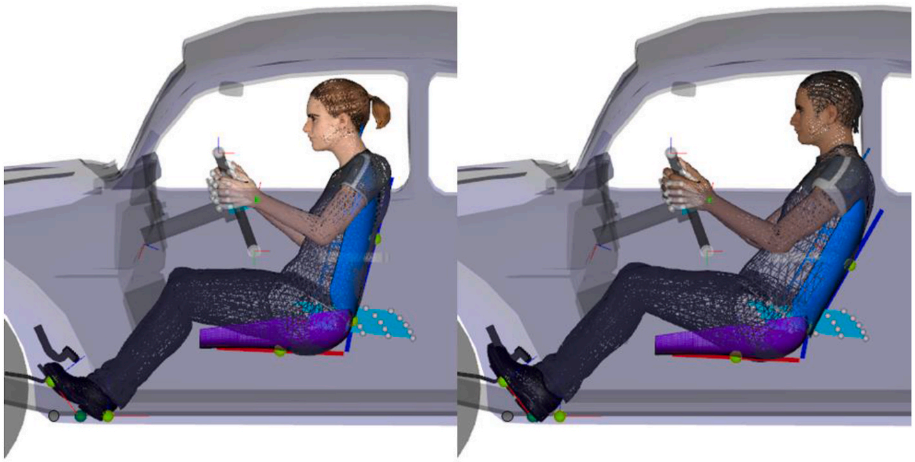

Fig. 8. Design scenario results visualisation: Manikin 1 – Adjustment 16(15°); Manikin 2 – Adjustment 21(30°). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

still includes actual design variables that are important to consider in vehicle ergonomics design processes, and represents critical aspects of driver accommodation. The ranges of the design variables are exaggerated to cover a larger area than usual. However, the range of the torso angle would have been more realistic with a smaller range, e.g. from 5° to 25° with 10° increments, instead of 0°–30° with 15° increments. Still, it is important to emphasise that the SBMOO approach can be extended to address various design challenges within occupant packaging, and by using other methods and evaluation metrics (Gao et al., 2021, 2022). In this case, the regression models in Park et al. (2016) were selected to have a posture evaluation method that considers anthropometric and postural variability across vehicle occupants. However, the regression models in Park et al. (2016) do not include any joint angle prediction equation for the upper limbs, which could be a limitation when predicting driver postures and using these prediction models to evaluate manikin postures, as in this study.

Despite the advantages and promising potentials of using the SBMOO approach for occupant packaging design, it is important to highlight that some specific challenges and limitations are closely tied to the capabilities of the DHM tool used. Current DHM tools provide a wealth of functions for predicting human posture and motion, but no contemporary DHM tool adequately considers or predicts all relevant aspects of human behaviour and form. For instance, the skeleton definition and body shape of the manikins in the version of IPS IMMA used in this study do not fully represent the diversity of potential human deformations that exist during real life driving scenarios. Moreover, cognitive factors, e.g. decision-making and mental workload, cannot be included in predictions because they are not a feature included in the predictive model used by IPS IMMA (Hanson et al., 2019; IPS IMMA, 2023). While these limitations are inherent in the software used, they also reflect broader challenges in the field of DHM (Demirel et al., 2021). Moreover, dealing with numerous variables and complex scenarios requires significant computational resources, which can also present challenges in terms of scalability and accessibility.

In summary, this study investigates using a SBMOO approach in occupant packaging design, aimed at finding design solutions that optimize driver postures for a diverse end-user group. While our method demonstrates efficacy, its efficiency, specifically in terms of time to complete an SBMOO analysis, has not been evaluated in this study and could be considered in future work. Future research challenges include comparing and evaluating the proposed method to currently used methods and exploring alternative optimization algorithms that may provide improved efficiency without compromising the quality of results. The proposed method could offer an alternative to more manual current approaches, potentially reducing the effort of engineers such as the challenges involved with manual methods (Perez Luque et al., 2022a). Further, the comparison between occupant packaging design tasks with current methods could be done by measuring factors like engineering man-hours. Additionally, although this study focuses on driver postures, different design scenarios tailored to passengers or diverse vehicles could be tested, such as trucks with unique occupant packaging requirements like stair-climbing, pedal placement, adjusting seat dimensions, or configuring control interfaces. Further research could also explore dynamic and time-based dependent evaluations, extending the application of SBMOO to larger manikin families and diverse vehicle types. This would enhance the accommodation of more diverse end-user groups and further increase the effectiveness of the work conducted by design engineers and ergonomics designers in the occupant packaging design process.

In conclusion, this study investigated and demonstrated the efficacy of using a SBMOO method combined with DHM tools to improve the effectiveness of occupant packaging design. The findings highlight the potential of SBMOO to assist ergonomics designers in balancing multiple conflicting objectives while considering human anthropometric and postural variability. These contributions provide valuable insights and a robust foundation for advancing the occupant packaging design process, thereby ultimately enhancing comfort and safety for diverse user populations.

CRediT authorship contribution statement

Estela Perez Luque: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aitor Iriondo Pascual:** Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Dan Högberg:** Writing – review & editing, Supervision. **Maurice Lamb:** Writing – review & editing, Supervision, Methodology. **Erik Brolin:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ergon.2024.103690.

Data availability

Data will be made available on request.

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